IMPLEMENTATION OF HIGH SPEED ENERGY EFFICIENT 4-BIT BINARY CLA BASED INCREMENTER DECREMENTER

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***Abstract*-- The paper presents the implementation of a high speed energy efficient 4-bit binary CLA based incrementer decrementer. The design methodology is extensively based on static CMOS logic and transmission gate logic to achieve higher operating frequencies, smaller delays and optimized area. This circuit is especially suitable for long bit incrementer/decrementer that can be used in program counter, frequency dividers and address generation unit in microprocessors. Simulation results illustrates that the designed adder has superior performance compared to existing adders in terms of power dissipation and speed. The proposed circuit is implemented on TSMC 0.18µm process model. The measurement results indicate that the proposed 4-bit incrementer/decrementer can operate up to 5GHz with 200x160 µm² optimized area.**

Keywords—Carry look-ahead adder(CLA), Carry ripple adder(CRA), INC/DEC(Incrementer/decrementor)

I. INTRODUCTION

Arithmetic Addition is one of the widely used fundamental arithmetic operations within contemporary electronic systems. Various energy efficient high speed adder topologies have been proposed to implement addition operation. Adder forms an essential component of many VLSI systems such as digital signal processors (DSP), microprocessors, ALUs, floating point arithmetic units, memory addressing units, program counters [1], frequency dividers [2] and MAC units. Such high performance devices need low power and area efficient adder circuits.

The choice of adder architecture is of utmost importance, since its performance determines the whole system response [3]. For the implementation of fast adder, CLA principle is more optimized as it consumes less amount of time duration to calculate final carry bit. However this high-speed design will spend more silicon area. On the other hand a high-speed parallel adder should be utilized in the adder-based incrementer / decrementer to improve the operating speed which can count up or count down by one step in one clock cycle. When the incrementer/decrementer has a long bit-width the operating speed may be degraded significantly.

Several architectures for incrementer/decrementer have been proposed for speed and power improvement. In [4], a ripple carry adder (RCA) based incrementer/decrementer has been discussed which has speed limitation due to carry propagation. Whereas in [5], the CLA based incrementer/decrementer circuit has been proposed in order to improve the speed. In [6], a MUX-based binary incrementer/decrementer circuit is designed which have more efficient in terms of both speed and hardware complexity as compared to RCA incrementer/decrementer.

This paper presents the implementation of a CLA based incrementer/decrementer using static CMOS logic and transmission gate logic. The proposed implementation lays emphasis on the usage of hybrid logic, i.e. coupling of transmission gates with CMOS gates at appropriate stages rather than using complete CMOS or transmission gate designs [4]. This reduces high power dissipation due to cascading of transmission gates while giving lesser propagation delay.

II. OVERVIEW OF BAISC MODULE OF CLA BASED INCREMENTER DECREMENTER

The aim of the designed digital circuit is to add/sub two 4 – bit unsigned binary numbers using the "Carry Look Ahead" logic.

Generally parallel prefix adder structure employs three main parts: the propagation/ generation block, sum generator, and carry generator. These parts of the circuit decide the overall circuit performance, power consumption, speed, area, power delay product etc. Extensive research has been done to improve the performance of these basic building blocks to design a large carry adder. Different type of architectures, multidimensional dynamic programming paradigm, logic families, algorithms have been presented to improve the performance of adder in terms of speed, power, energy, delay, noise etc.

The generate / propagate block, sum generator and carry generator can be realized using the expression (1)-(8) [7].

$Gi = Ai.\ Bi$ ---------- (1) Carry generate.

$Pi = Ai \oplus Bi$ --------- (2) Carry propagate.

Both propagate and generate signals depend only on the input bits and thus will be valid after one gate delay. The

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new expressions for the output sum and the carryout are given by:

$$S_i = P_i \oplus C_i \text{---------- (3)}$$

$$C_{i+1} = G_i + P_i.C_i \text{ ----- (4)}$$

The final equations of 4-bit CLA adder are defined as:

$$C1 = G0 + P0C0 \text{---------- (5)}$$

$$C2 = G1 + P1C1 = G1 + P1(G0 + P0C0)$$
$$= G1 + P1G0 + P1P0C0 \text{---------- (6)}$$

$$C3 = G2+P2C2 = G2 + P2G1 + P2P1G0 + P2P1P0C0 \text{-- (7)}$$

$$C4 = G3 + P3C3 = G3 + P3G2 + P3P2G1 + P3P2P1G0$$
$$+ P3P2P1P0C0 \text{---------- (8)}$$

To realize 4-bit CLA based INC/DEC, five D Flip-Flop (Fig1) are used to synchronize INC/DEC with the positive clock edge, four propagation/ generation blocks (Fig2) to generate the propagation/generation bits to operate CLA, 4-bit carry generator circuit (Fig3) and four sum generator block (Fig4) which will give the final output of CLA based INC/DEC. The transistor level circuit diagram of all these sub-modules are shown below as per the mentioned equations.

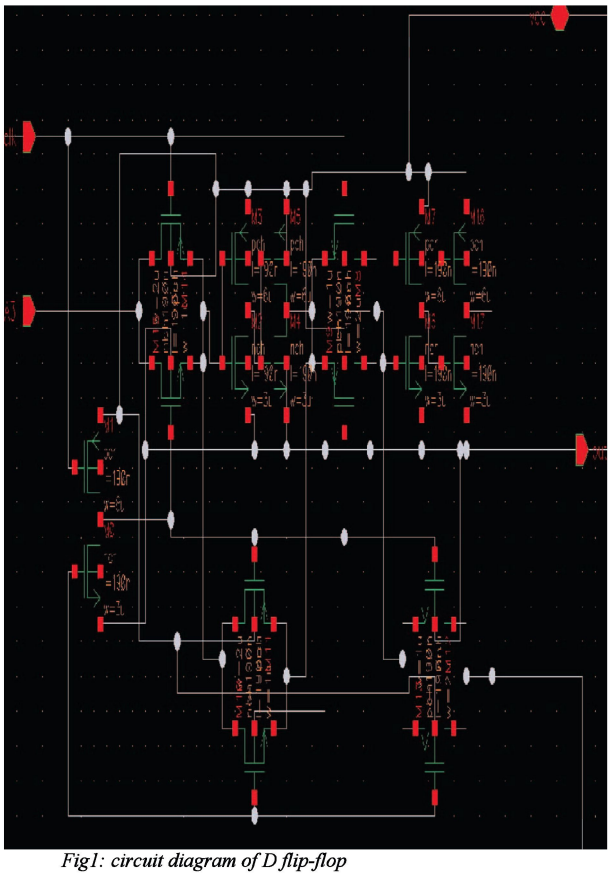


Fig1: circuit diagram of D flip-flop

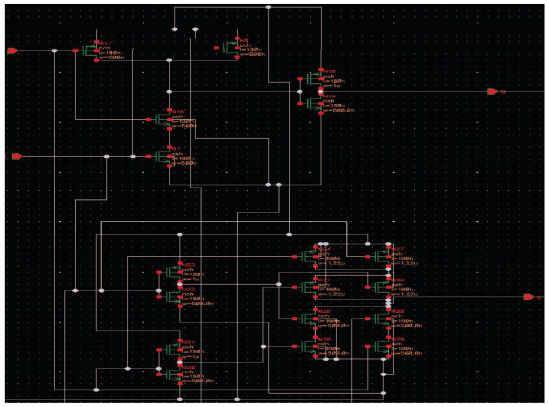


Fig2: circuit diagram of propagation / generation block

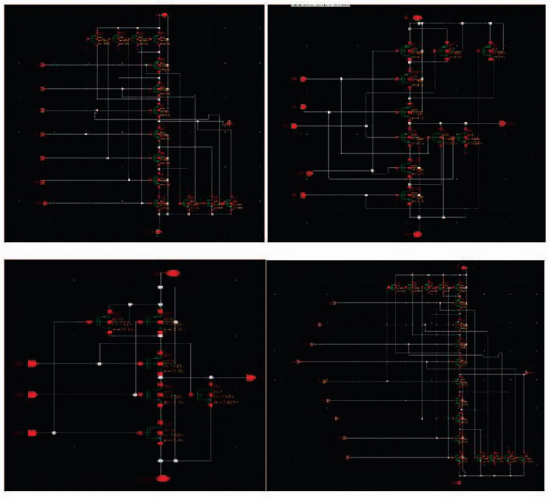


Fig3: Circuit diagram of 4-bit carry generator C1, C2, C3, C4

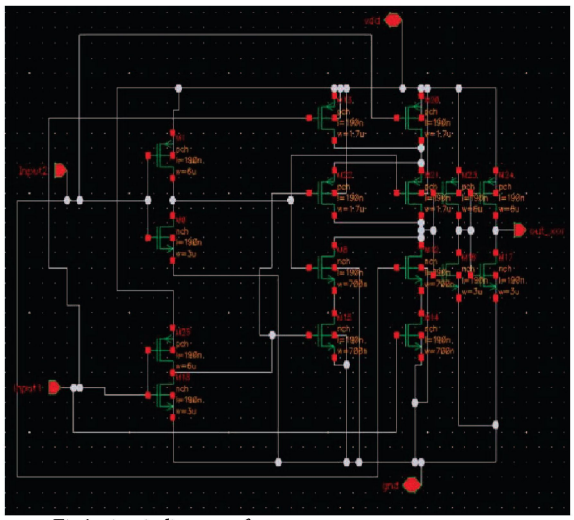


Fig4: circuit diagram of sum generator

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III. PROPOSED ARCHITECTURE FOR CLA BASED INCREMENTER DECREMENTER

The proposed 4-bit CLA-based INC/DEC is shown in Fig.5. The first 4-bit input Ai propagates through a D flip-flop to synchronize INC/DEC with the positive clock edge while the second 4-bit input Bi which is XORed with input carry bit (C0) , the output of both modules goes into propagation / generation blocks and the output of this block goes as the input of the carry generation unit. The carry bits are XORed with respective propagation bits to give the final output bits of the CLA as shown in Fig5.

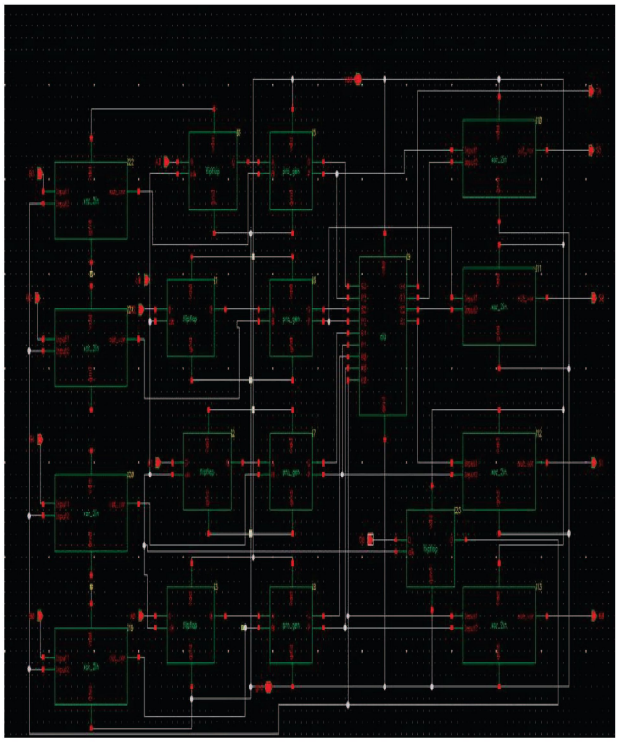


Fig5: circuit diagram of CLA based INC/DEC

The function of CLA based INC/DEC mainly depends on the value of C0 when C0= 0 it will increment the corresponding inputs and vice versa. If the function to be implemented is decrement, all the input bits should be inverted first and then sent to the input of the circuit. The following example is used to demonstrate the operation of the proposed incrementer / decrementer. Assume the 4-bit inputs are 1011 and 1111 with carry bit i.e. C0 =0/1 with the least significant bit (LSB) placed at the rightmost position and the most significant bit (MSB) placed at the leftmost position. The output of the CLA will be 11010/ 11100 (INC/DEC) respectively.

IV. PHYSICAL LAYOUT DESIGN OF CLA BASED INCREMENTER DECREMENTER

Physical layout design affects the overall circuit performance (area, speed and power dissipation) since the physical structure determines the transconductance of the transistors, the parasitic capacitances and resistances, and obviously, the silicon area which is used to realize a certain function. Fig (6)-Fig (9) shows the layout of basic modules to implement 4-bit CLA based INC/DEC. Fig (10) shows the final layout of CLA based Incrementer Decrementer. The size of the final layout is 200x160 μm^2.

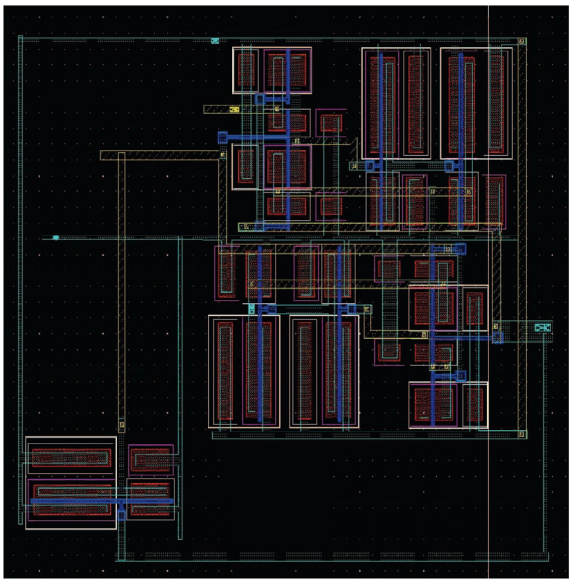


Fig6: Layout of D flip-flop

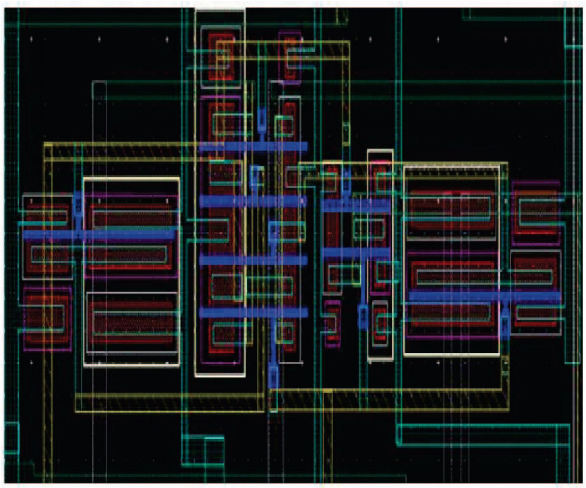


Fig7: Layout of propagation/generation block

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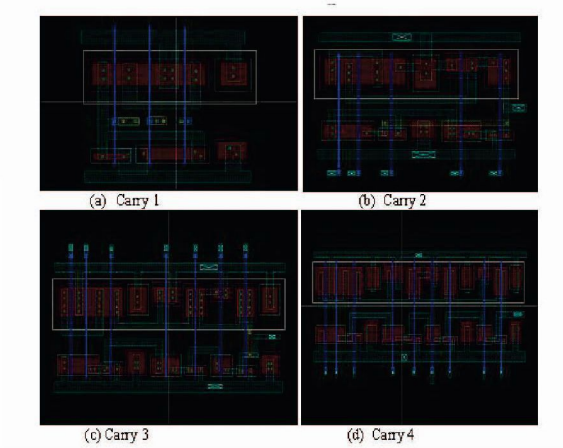


Fig8: layout of carry generation block

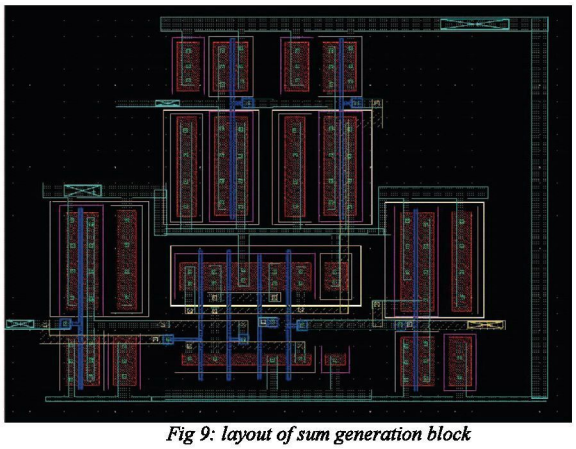


Fig 9: layout of sum generation block

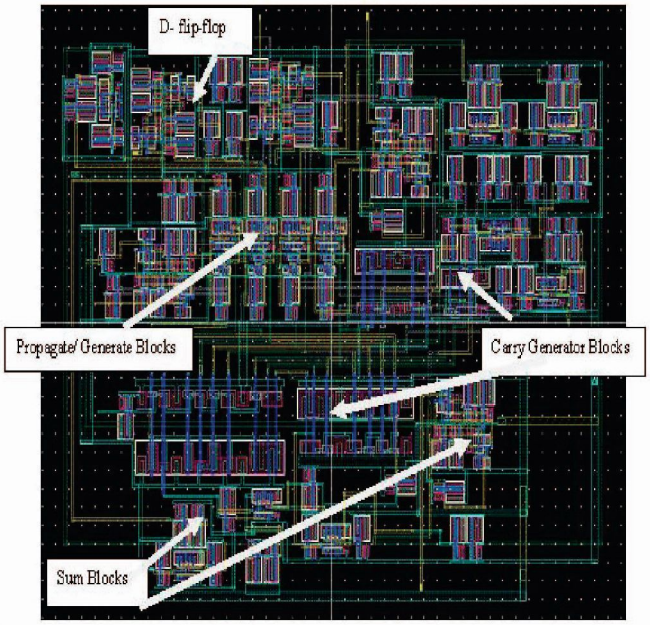


Fig 10: Layout of CLA based Incrementer Decrementer

V. PERFORMANCE EVALUATION AND EXPERIMENTAL RESULTS

A parametric analysis on varying the PMOS transistor width is done to observe the power consumption. The power consumption is least when we have minimum size for both NMOS and PMOS. The W/L's of all transistors in the sub-systems were kept in the ratio of 2:1 for the pull up and pull down network respectively. Table 1 summarizes the various transistor sizes used for the various sub-systems used in our circuit design. For all transistors the L was fixed at 190nm.

Table 1: Calculation of W/L's of various sub - systems

Sr. No	Sub - System	W of Pull Up Network (in micro-meters)	W of Pull Down Network (in micro-meters)	Ratio of Pull Up: Pull Down Network
1.	Inverter (NOT)	6	3	2 : 1
2.	AND - 2 Input	.7	.35	2 : 1
3.	XOR - 2 Input	.85	.35	2.42 : 1
4.	Carry 1	1.4	.7	2 : 1
5.	Carry 2	1.4	.7	2 : 1
6.	Carry 3	1.4	.7	2 : 1
7.	Carry 4	1.4	.7	2 : 1
8.	D - Flipflop (Transmission gates)	2	1	2 : 1

The pre-layout and post-layout simulation results are shown at SS mode in Fig. 11 and Fig. 12 respectively. The results are for the following inputs:
A = 1111/0000 changing at a frequency of 100MHz
B =B3 B2 B1 B0; B3, B2 = 11/00 changing at a frequency of 25MHz
B1, B0 = 11/00 changing at a frequency of 100MHz
C0 = 0/1 changing at a frequency of 50 MHz
Clk = 1/0 changing at a frequency of 5GHz.

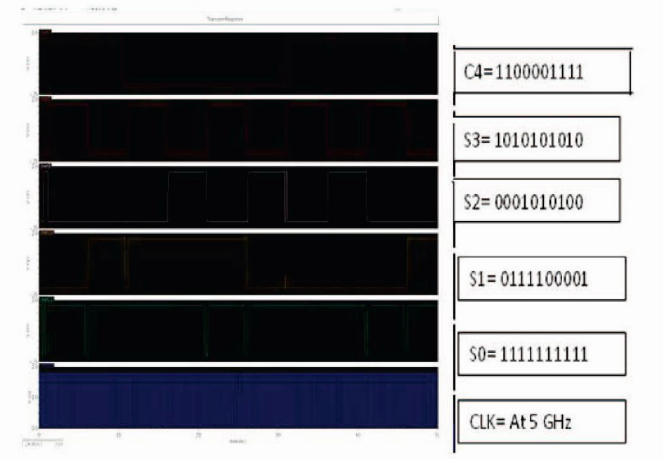


Fig11: pre-layout simulation results at SS mode

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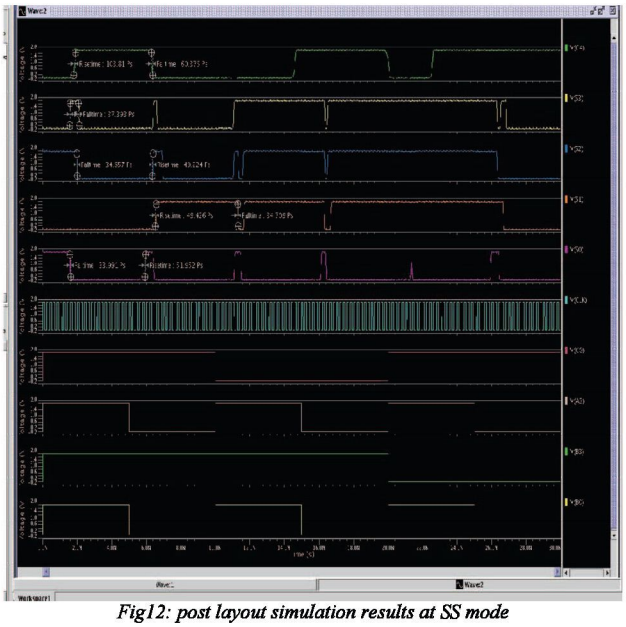


Fig12: post layout simulation results at SS mode

The results of schematic simulation along with the design specs are tabulated and compared with the results obtained by post-layout simulations in table 2.

Table 2: Results

Output	Schematic Simulation					Post - Layout Simulations				
	Rise time (ps)	Fall Time (ps)	t_{PHL} (ps)	t_{PLH} (ps)	Delay (ps)	Rise time (ps)	Fall Time (ps)	t_{PHL} (ps)	t_{PLH} (ps)	Delay (ns)
C4	260	203.7	174.3	212.6	193.45	106.31	59.103	69.08	280.57	1.659
S3	162.5	115	72.1	95	83.55	50.375	35.494	38.873	50.402	1.44
S2	160	113.3	73.5	99	86.25	49.604	38.696	34.914	46.509	1.422
S1	161	115.9	70	90.6	80.3	49.415	34.916	37.793	46.45	0.9866
S0	167	166	68.9	96	164.9	51.946	36.394	41.121	188.87	1.599
Avg. Power Dissipation			2.971 mW			Avg. Power Dissipation			5.2072 mW	
Total Power			9.42 mW			Total Power			7.497 mW	
Power - Delay Product (PDP)			1.818 pJ			Power - Delay Product (PDP)			12.437 pJ	
Total No. of Transistors			362			Total No. of Transistors			362	
Max. Frequency			5 GHz			Max. Frequency			5 GHz	

VI. CONCLUSION

A 4 - bit CLA based binary incrementer/decrementer was designed and various post-layout simulations were carried out to check the functionality of the system. Beginning with the designing phase the classical CMOS design methodology was used so as to get the required design specification of achieving high speeds and lesser delays. To obtain more prominent results the sub module D flip-flop is implemented using the Transmission gate CMOS methodology.

After performing post schematic simulations it was obtained that the desired specifications of achieving high speeds and lesser delays was satisfied and the circuit with 362 transistors was operational with a total power consumption of 9.42mW at a frequency of 5GHz and the delays being in the range of 0-100ps.

The main feature being that the maximum clock speed was same after the post schematic simulations - 5GHz. The power obtained after the post - layout simulations was 7.5mW and the delays were in the range of 0-1.5ns. The area occupied by the layout is 200x160 um^2.

Thus we implemented a high speed 4 -bit CLA based binary incrementer/decrementer.

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